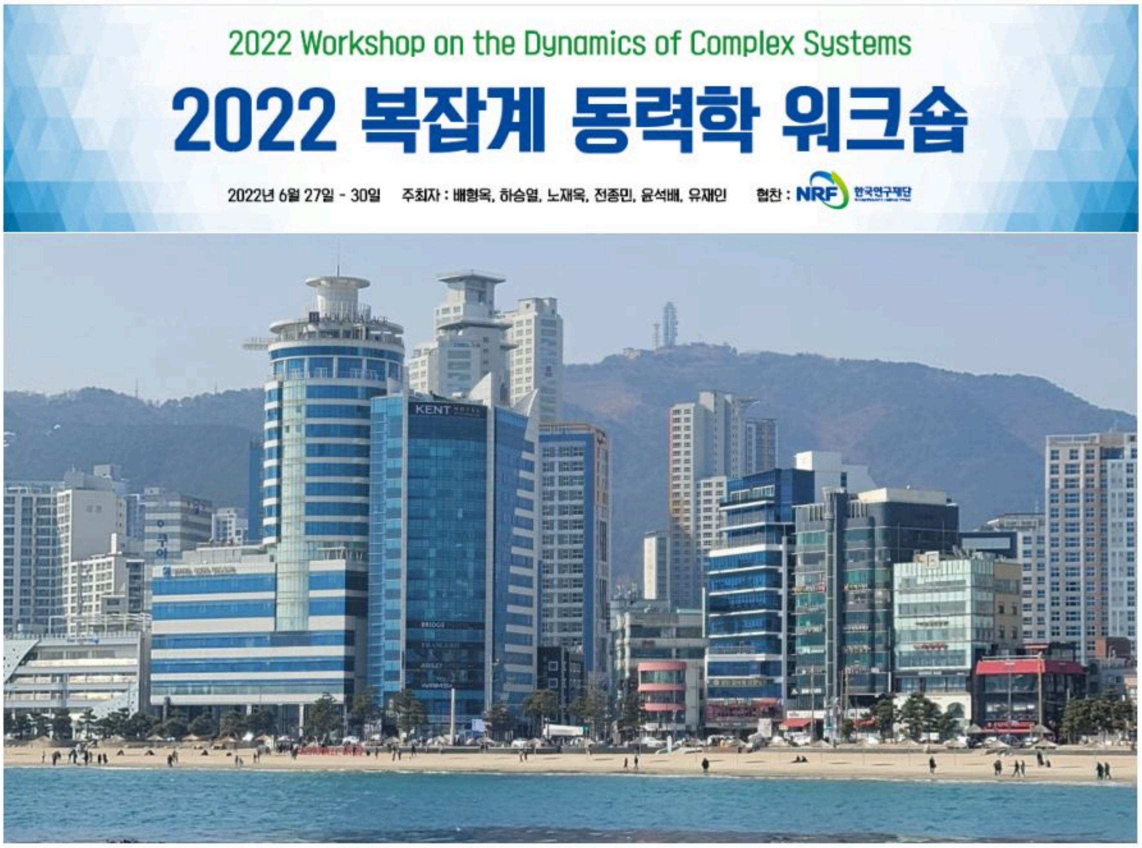
2022 Workshop on the Dynamics of Complex Systems

(Modeling, Analysis and Application)

Haeundae Central Hotel, Busan, Korea

June 27(Mon) ~ June 30(Thu), 2022

Program and Abstracts



Date: 27(Mon) - 30(Thu), June, 2022

Conference Venue: Haeundae Central Hotel, Busan, Korea

Organizers:
Hyeong-Ohk Bae (Ajou University)
Seung-Yeal Ha (Seoul National University)
Jongmin Jeon (Busan)
Jaiok Roh (Hallym University)
Jane Yoo (Ajou University)
Seok-Bae Yun (Sungkyunkwan University)

Sponsors:
National Research Foundation of Korea
Ajou University
Hallym University
Seoul National University
Sungkyunkwan University

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Invited Talks

(June 27th)

June 27 (Monday)

13:00 - 14:00	Registration
14:00 – 14:25	**Yong-Hoon Lee** A constructive approach about the existence of positive solutions for Minkowski curvature problems
14:25 – 14:50	**Minkyu Kwak** A remark on the existence of an inertial manifold
14:50 – 15:00	Coffee Break
15:00 – 15:25	**Jongmin Han** Asymptotics for n-component Ginzburg-Landau vortices
15:25 – 15:50	**Sang-Il Han** Affinity maturation of Covid 19 Spike protein and Ace2
15:50 – 16:00	Coffee Break
16:00 – 16:25	**Jinwook Jung** The pressureless damped Euler-Riesz equations
16:25 – 16:50	**Jeongho Kim** Rigorous derivation of the Euler-alignment model with singular communication weights from a kinetic Fokker-Planck-alignment model

A constructive approach about the existence of positive solutions for Minkowski curvature problems

Yong-Hoon Lee

June 27, 2022

Department of Mathematics, Pusan National University
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Abstract

In this paper, we give an existence theorem about positive solutions for the Dirichlet boundary value problem of one dimensional Minkowski curvature equations. We apply the theorem to one parameter family of problems to investigate a constructive method for numerical range of parameters where positive solutions exist. Moreover, we establish a nonexistence theorem of positive solutions for the corresponding one parameter family of problems. The coefficient function may be singular at the boundary and nonlinear term satisfies a sublinear growth condition. Main argument for the proof of existence theorem is employed by Krasnoselskii's theorem of cone expansion and compression. We give a numerical algorithm and various examples to illustrate numerical information about ranges of the existence and nonexistence parameters which have been given only in a theoretical manner so far.

Memo

A remark on the existence of an inertial manifold

Minkyu Kwak

June 27, 2022

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Abstract

The inertial manifold theory has been developed in the mid- 80's. However, the theory, even after enormous efforts, does not apply to the Navier-Stokes equations simply because the spectral gap condition is not satisfied. Here we focus on backward bounded (in time) solutions for the system and introduce a new insight to relax the gap condition.

Memo

Asymptotics for n-component Ginzburg-Landau vortices

Jongmin Han

June 27, 2022

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Abstract

Let $\Omega \subset \mathbb{R}^2$ be a smooth bounded simply connected domain. In this talk, we consider the n-component Ginzburg-Landau equations on Ω:

$$\begin{cases} -\Delta u_i = \dfrac{1}{\varepsilon^2} u_i \Big(n - \displaystyle\sum_{j=1}^{n} |u_j|^2 \Big) \quad \text{in} \quad \Omega, \\ \quad u_i = g_i \quad \text{on} \quad \partial\Omega, \end{cases}$$

for each $i = 1, \cdots, n$. Here, $u_1, \cdots, u_n : \Omega \to \mathbb{R}^2$, and $g_1, \cdots, g_n : \partial\Omega \to S^1$ are smooth maps such that

$$\deg(g_j, \partial\Omega) = d_j \in \mathbb{N} \cup \{0\} \quad \text{for} \quad j = 1, \cdots, n.$$

We study the asymptotic behavior of minimizers $(u_{1,\varepsilon}, \cdots, u_{n,\varepsilon})$ for the energy functional

$$E_{\varepsilon,\Omega}(u_1, \cdots, u_n) = \frac{1}{2} \int_{\Omega} \sum_{j=1}^{n} |\nabla u_j|^2 dx + \frac{1}{4\varepsilon^2} \int_{\Omega} \Big(n - \sum_{j=1}^{n} |u_j|^2 \Big)^2 dx$$

on $H^1_{g_1}(\Omega, \mathbb{R}^2) \times \cdots \times H^1_{g_n}(\Omega, \mathbb{R}^2)$. We prove that the minimizers converges locally in any C^k-norm to a solution $(u_1^*, \cdots, u_n^*)$ of a system of generalized harmonic map equations

$$\begin{cases} -\Delta u_j = \dfrac{1}{n} u_j \displaystyle\sum_{k=1}^{n} |\nabla u_k|^2 \quad \text{in} \quad \Omega, \\ \quad u_j = g_j \quad \text{on} \quad \partial\Omega, \\ \displaystyle\sum_{j=1}^{n} |u_j|^2 = n \quad \text{in} \quad \Omega. \end{cases}$$

Memo

Affinity maturation of Covid 19 Spike protein and Ace2

Sang-Il Han

June 27, 2022

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Abstract

In order to respond to pandemic Covid 19, which has spread worldwide after 2020, mankind is responding by developing mRNA Vaccine and therapeutics. Vaccines are treated enhancing the human immune system, and the therapeutics are structured to treat by interfering with the replication of the virus. All of these methods are performed after the virus infects the cells. However, since Covid 19 is a structure that infects lung cells by reacting with spike protein and Ace2, it will be useful to accurately identify the reaction between the two proteins. Therefore, this study aims to investigate the phenomenon that affinity maturation occurs in quantum chemistry after constructing molecular dynamics between two proteins for 500 ns. For the above study, we intend to use Gromacs for molecular dynamics and Cp2k for bonding analysis.

Memo

The pressureless damped Euler-Riesz equations

Jinwook Jung

June 27, 2022

Research Institute of Basic Sciences, Seoul National University
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Abstract

In this talk, we consider the pressureless damped Euler–Riesz equations posed in either $\mathbb{R}^d$ or $\mathbb{T}^d$. We construct the global-in-time existence and uniqueness of classical solutions for the system around a constant background state. We also establish large-time behaviors of classical solutions showing the solutions towards the equilibrium as time goes to infinity. For the whole space case, we first show the algebraic decay rate of solutions under additional assumptions on the initial data compared to the existence theory. We then refine the argument to have the exponential decay rate of convergence even in the whole space. In the case of the periodic domain, without any further regularity assumptions on the initial data, we provide the exponential convergence of solutions. This talk is based on the joint work with Young-Pil Choi (Yonsei University).

Memo

Rigorous derivation of the Euler-alignment model with singular communication weights from a kinetic Fokker-Planck-alignment model

Jeongho Kim

June 27, 2022

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Abstract

We present a rigorous derivation of the isothermal Euler-alignment model with singular communication weights. We consider a hydrodynamic limit of a kinetic Fokker-Planck-alignment model, which is the nonlinear Fokker-Planck equation with the Cucker-Smale alignment force. Our analysis is based on the estimate of relative entropy between macroscopic quantities, together with careful analysis on the singular communication weights.

Memo

Invited Talks
(June 28th)

June 28 (Tuesday)

10:00 - 10:25	**Namkwon Kim (Chosun University)** Remark for multiplicity for stationary Navier-Stokes equations
10:25 - 10:50	**Young-Sam Kwon (Dong-A University)** Introduction to dissipative turbulent solutions of compressible models and applications
10:50 - 11:00	Coffee Break
11:00 - 11:25	**Seok-Bae Yun (Sungkyunkwan University)** BGK model of the Boltzmann equation and its variants
11:25 - 11:50	**Jane Yoo (Ajou University)** Online Portfolio Optimization based on Consensus Based Optimization Algorithm with Adaptive Momentum and Average Drift
11:50 - 14:00	Lunch
14:00 - 14:25	**Seung-Yeon Cho (Gyeongsang National University)** Local velocity grid conservative semi-Lagrangian schemes for the BGK model
14:25 - 14:50	**Dongnam Ko (Catholic University of Korea)** On the stochastic synchronization of the Winfree model with a multiplicative noise
14:50 - 15:00	Coffee Break
15:00 - 15:25	**Gi-Chan Bae (Seoul National University)** BGK model for multi-component gases near a global Maxwellian
15:25 - 15:50	**Woojoo Shim (KIAS)** Stochastic clustering dynamics of a nonlinear consensus model
15:50 - 16:00	Coffee Break
16:00 - 16:25	**Myeongju Kang (Seoul National University)** Continuum limit of Lohe model on some class of Lie groups
16:25 - 16:50	**Hangjun Cho (Seoul National University)** Cardinality of collisions in the asymptotic phase-locking for the Kuramoto model with inertia
17:00 - 19:00	Banquet

Remark for multiplicity for stationary Navier-Stokes equations

Namkwon Kim

June 28, 2022

Department of Mathematical Sciences, Chosun university
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Abstract

We consider the incompressible stationary Navier-Stokes equations in a bounded domain. It is well-known that the solutions exist for a given data under a mild condition for the data. However, it is not rigorously stated yet whether the solution is unique or not. We present some conditions and examples for the multiplicity.

Memo

Introduction to dissipative turbulent solutions of compressible models and applications

Young-Sam Kwon

June 28, 2022

Department of Mathematics, Dong-A University
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Abstract

We will introduce the notion of dissipative turbulent solution of compressible models which is very weak solution to compressible NSE. We also introduce a long-time behaviour and asymptotic limit problems of this solutions. Finally, we will suggest open problems.

Memo

BGK model of the Boltzmann equation and its variants

Seok-Bae Yun

June 28, 2022

Department of Mathematics, Sungkyunkwan University
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Abstract

The Boltzmann equation is a fundamental model that connects the particle regime and the fluid regime of rarefied gases. However, the application of the Boltzmann equation to various flow problems has been severely restricted by the intricate structure of the collision operator that requires severe computational costs. The observation made by Bhatnagar, Gross, and Krook in their attempts to overcome this difficulty is that the relaxation process can successfully describe the complicated process of collision after a short time scale. The equation introduced based on this observation is now called the BGK model, which has enjoyed great popularity as a numerically amenable equation that provides qualitatively satisfactory results. In this talk, we overview recent progress in the mathematical study of the BGK model and its various modifications.

Memo

Online Portfolio Optimization based on Consensus Based Optimization Algorithm with Adaptive Momentum and Average Drift

Jane Yoo

June 28, 2022

Department of Financial Engineering, Ajou University
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Abstract

Tracing a moving objective in a changing high dimensional space is prevalent but difficult optimization problem. We propose an applied consensus-based optimization algorithm with adaptive momentum estimation and average drift (Adamad-CBO) for tracking a moving optima. We show the convergence to the global minimizer of the Adamad-CBO and verify the algorithm's performance in approximating both static and dynamic objective function. In our financial applications, the Adamad-CBO improves the speed of searching, while lowering tracking errors than other CBO methods in finding the moving optima in the online portfolio optimization. This is the joint work with Hyeong-Ohk Bae, Seung-Yeal Ha, Chanho Min, and Jaeyoung Yoon.

Memo

Local velocity grid conservative semi-Lagrangian schemes for the BGK model

Seung-Yeon Cho

June 28, 2022

Department of Mathematics, Gyeongsang National University
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Abstract

Most numerical schemes proposed for solving kinetic equations for rarefied gas dynamics, such as Boltzmann equation and BGK model, are based on the discrete velocity approximation. In general, such approaches use fixed velocity grids, and one must secure a sufficient number of grid points in phase space to resolve the structure of the distribution function. However, when dealing with the problem with large variation of mean velocity and temperature, the computational cost and memory allocation requirements become prohibitively large.

In this talk, we introduce a velocity adaption technique for the semi-Lagrangian scheme applied to the BGK model. The velocity grids will be set locally in time and space. We apply a weighted minimization approach to impose global conservation and improve positivity, generalizing the L^2-minimization technique introduced by I. Gamba et al. We demonstrate the efficiency of the proposed scheme in several numerical examples. This is the joint work with G. Russo and S. Boscarino.

Memo

On the stochastic synchronization of the Winfree model with a multiplicative noise

Dongnam Ko

June 28, 2022

Department of Mathematics, The Catholic University of Korea
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Abstract

The long-time behaviors of stochastic oscillators mainly follows the Fokker-Planck equations, however, the detailed particle dynamics could show a kind of non-intuitive behaviors. Our focus is mainly on the stochastic Winfree oscillators under the multiplicative white noise. The additive white noise affects the dynamics as with the heat kernel, which makes the system spread its phases into the whole domain, where the ensemble cannot gather into equilibrium points. On the other hand, with the multiplicative noise, the individuals march toward regions with small noise strength, as in the geometric Brownian motion. In this talk, we establish the emergence of synchronization for the identical stochastic Winfree model. Here the main ingredient is the stochastic Barbalat's lemma, which shows the convergence of the potential or other energy functions. This analysis can be directly applied to the general gradient system with multiplicative noise. This is the joint work with Seung-Yeal Ha and Woojoo Shim.

Memo

BGK model for multi-component gases near a global Maxwellian

Gi-Chan Bae

June 28, 2022

Research institute of Mathematics, Seoul National University
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Abstract

In this talk, we establish the existence of the unique global-in-time classical solutions to the multi-component BGK model when the initial data is a small perturbation of global equilibrium. The main difference between this model compared to the classic mixture Boltzmann equation is that the rate of interchange of momentum and energy between the two species is controllable through two free parameters.

Based on the energy method, we carefully analyze the dissipative nature of the linearized multi-component relaxation operator, and observe that the partial dissipation from the intra-species and the inter-species linearized relaxation operators are combined in a complementary manner to give rise to the desired dissipation estimate of the model. Furthermore, the dissipation estimate depends on the momentum interchange rate and the energy interchange rate. So the larger interchange rate leads to stronger dissipation, and therefore, the faster convergence to the global equilibrium. This is joint work with C. Klingenberg, M. Pirner, and S.-B. Yun.

Memo

Stochastic clustering dynamics of a nonlinear consensus model

Woojoo Shim

June 28, 2022

School of Mathematics, Korea Institute for Advanced Study
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Abstract

Emergence of multi-clusters is one of the novel dynamical features of the deterministic consensus model. Recently, a complete cluster predictability of the first-order nonlinear consensus model originated from Cucker-Smale dynamics on the real line has been studied, where the clustering algorithm explicitly determines the number of asymptotic clusters and their cluster velocities only in terms of the initial data, coupling strength and supremum of coupling function. In this talk, we present stochastic noise response on the complete cluster predictability when the deterministic consensus model is stochastically perturbed in a multiplicative way. Our analytic findings illustrate that the same algorithm can be used to predetermine the clustering dynamics, but only for generic values of system parameters that excludes critical values.

Memo

Continuum limit of Lohe model on some class of Lie groups

Myeongju Kang

June 28, 2022

Research Institute of Basic Sciences, Seoul National University
E-mail : **bear0117@snu.ac.kr**

Abstract

The Lohe model is a generalized high dimensional Kuramoto-type model whose oscillators lie on a special class of matrix Lie group. The continuum Lohe model is a time-evolutionary integro-differential equation which governs the Lohe phase field, and solution to the Lohe model becomes a simple function valued solution to the continuum Lohe model. In this talk, we study the emergent dynamics and global well-posedness of the continuum Lohe model which can be obtained by a continuum limit of the Lohe model. We first construct a local solution to the continuum Lohe model. Then, we find an invariant set to extend our local solution to a global one. Lastly, we show that sequence of simple functions obtained from the Lohe model converges to a solution of the continuum Lohe model in supreme norm sense. This talk is based on the joint work with Hangjun Cho and Seung-Yeal Ha.

Memo

Cardinality of collisions in the asymptotic phase-locking for the Kuramoto model with inertia

Hangjun Cho

June 28, 2022

Department of Mathematical Sciences, Seoul National University
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Abstract

We study the cardinality of collisions between Kuramoto oscillators in the (asymptotic) phase-locking process in the presence of inertia. In the absence of inertia, it has been known that the finiteness of collisions between oscillators is equivalent to the emergence of phase-locking. Thus, a natural question is whether this finiteness result is still valid for the Kuramoto model with inertia or not. In a small inertia regime, we show that the finiteness of collisions is also equivalent to phase-locking like the Kuramoto model. In contrast, in a large inertia regime, we show that homogeneous Kuramoto ensemble with the same natural frequency can reach phase-locking, while there are a countable number of collisions between oscillators. This is the contrasted effect of a large inertia in phase-locking process. This talk is based on the joint work with Jiu-Gang Dong and Seung-Yeal Ha.

Memo

Invited Talks

(June 29th)

June 29 (Wednesday)

10:00 - 10:25	**Dohyun Kim (Sungshin Women's University)** Asymptotic behavior for a system of coupled Schrödinger equations
10:25 - 10:50	**Doheon Kim (Hanyang University (ERICA))** A unified framework for distributed optimization algorithms over time-varying directed graphs
10:50 - 11:00	Coffee Break
11:00 - 11:25	**Junhyeok Byeon (Seoul National University)** Collective behaviors of second-order nonlinear consensus models with a bonding force
11:25 - 11:50	**Myeongu-Su Lee (Sungkyunkwan University)** Derivation of relativistic BGK model for gas mixtures
11:50 - 14:00	Lunch
14:00 - 17:00	Discussion

Asymptotic behavior for a system of coupled Schrödinger equations

Dohyun Kim

June 29, 2022

School of Mathematics, Statistics and Data Science, Sungshin Women's University
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Abstract

In this talk, we introduce a system for coupled Schrödinger equations, so-called Schrödinger-Lohe (SL) equation describing possible quantum synchronization. First, the recent progress on the SL equation with a specific viewpoint on its emergent dynamics is presented by following a recent review article. Then, we provide a state-of-the-art result for the SL equation with heterogeneous wave functions. Precisely, convergence toward equilibrium is established when the coupling is dominant. Our strategy mainly depends on the method of dimension reduction and the dynamics of the Kuramoto model. This talk is based on the joint work with Dr. Hansol Park (Simon Fraser University).

Memo

A unified framework for distributed optimization algorithms over time-varying directed graphs

Doheon Kim

June 29, 2022

Department of Applied Mathematics, Hanyang University (ERICA)
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Abstract

In this paper, we propose a framework under which several decentralized optimization algorithms can be treated in a unified manner. More precisely, we show that the distributed subgradient descent algorithms, the subgradient-push algorithm, and the distributed algorithm with row-stochastic matrix can be derived by making suitable choices of consensus matrices, step-size and subgradient from the decentralized subgradient descent. As a result of such unified understanding, we provide a convergence proof that covers the aforementioned algorithms under a novel algebraic condition that is strictly weaker than the conventional graph-theoretic condition. This unification also enables us to derive a new distributed optimization scheme.

Memo

Collective behaviors of second-order nonlinear consensus models with a bonding force

Junhyeok Byeon

June 29, 2022

Department of Mathematical Sciences, Seoul National University
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Abstract

We study the collective behaviors of two second-order nonlinear consensus models with a bonding force, namely the Kuramoto model and the Cucker-Smale model with inter-particle bonding force. The proposed models contain feedback control terms which induce collision avoidance and emergent dynamics in a suitable framework. Through the cooperative interplays between feedback controls, initial state configuration tends to an ordered configuration asymptotically under suitable frameworks which are formulated in terms of system parameters and initial configurations. For a two-particle system on the real line, we show that the relative state for the Cucker-Smale model with a bonding force tends to the preassigned value asymptotically, and we also provide several numerical examples to analyze the possible nonlinear dynamics of the proposed models, and compare them with analytical results. This is the joint work with Hyunjin Ahn, Seung-Yeal Ha and Jaeyoung Yoon.

Memo

Derivation of relativistic BGK model for gas mixtures

Myeongu-Su Lee

June 29, 2022

Department of Mathematics, Sungkyunkwan University
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Abstract

The BGK model is the best-known model equation of the Boltzmann equation, which satisfies the essential features of the Boltzmann equation at a much lower computational cost. In this talk, we present a relativistic BGK-type model for gas mixtures. And we address the determination problem of the auxiliary parameters constituting the attractor, and some properties that our model satisfies.

Memo

Invited Talks
(June 30th)

June 30 (Thursday)

10:00 - 10:25	**Euntaek Lee (Seoul National University)** Generalized aggregation model on Hilbert C^*-modules
10:25 - 10:50	**Gyuyoung Hwang (Seoul National University)** On the complete aggregation of the Wigner-Lohe model for identical potentials
10:50 - 11:00	Coffee Break
11:00 - 11:25	**Sang-Yun Nam (Ajou University)** Solving Black-Scholes PDE associated with local volatility via Physics-Informed Neural Network
11:25 - 11:50	**Seung-Gu Kang (Ajou University)** Black-Scholes model associated with Local Volatility
11:50 - 14:00	Lunch
14:00 - 17:00	Discussion

Generalized aggregation model on Hilbert C^*- modules

Euntaek Lee

June 30, 2022

Department of Mathematical Sciences, Seoul National University
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Abstract

We present a new aggregation model on Hilbert C^*-modules and study its emergent dynamics. Hilbert C^*-module is a generalized notion of Hilbert space by allowing the inner product to take values in a C^*-algebra rather than $\mathbb{C}$. The most simple form of our model encompasses various Kuramoto-type aggregation models that already exist, such as the Kuramoto model, the swarm sphere model, the Lohe matrix model, the group ring model, the Schrödinger-Lohe model, the complex sphere model in Hilbert space, Lohe tensor model and the Stiefel manifold model. For the proposed model, we provide several sufficient criteria leading to the asymptotic aggregation of particles. We also compare our new estimate with previous results to verify its optimality. This is the joint work with Seung-Yeal Ha and Woojoo Shim.

Memo

On the complete aggregation of the Wigner-Lohe model for identical potentials

Gyuyoung Hwang

June 30, 2022

Department of Mathematical Sciences, Seoul National University
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Abstract

We study the collective behaviors of the Wigner-Lohe (WL) model for quantum synchronization in phase space which corresponds to the phase description of the Schrödinger-Lohe (SL) model for quantum synchronization, and it can be formally derived from the SL model via the generalized Wigner transform. For this proposed model, we show that the WL model exhibits asymptotic aggregation estimates so that all the elements in the generalized Wigner distribution matrix tend to a common one. On the other hand, for the global unique solvability, we employ the fixed point argument together with the classical semigroup theory to derive the global unique solvability of mild and classical solutions depending on the regularity of initial data. This is a joint work with Seung-Yeal Ha and Dohyun Kim.

Memo

Solving Black-Scholes PDE associated with local volatility via Physics-Informed Neural Network

Sang-Yun Nam

June 30, 2022

Department of Financial Engineering, Ajou University
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Abstract

In this article, we solve the Black-Scholes Partial Difference Equation(BS PDE) under the local volatility model by using an artificial neural network, and obtain the price and the greeks of derivatives in forms of function simultaneously.

Monte Carlo Method and Finite Difference Method are most well-known numerical methods for option pricing. However, they have some disadvantages. Typically, price and greeks cannot be obtained at the same time. the mesh-free point collocation method is used to calculate option price and their greeks simultaneously. Neural networks can be used in a sense similar to the mesh-free methods, which compute price and greeks simultaneously. It can also solve parametric PDEs and approximate even the derivatives with respect to the parameter

Memo

Black-Scholes model associated with Local Volatility

Seung-Gu Kang

June 30, 2022

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Abstract

In the financial market, an option is one of financial derivatives, which has the right to buy(call) or sell(put) by a specified date(maturity) for a specified price(strike or exercise price). Fischer Black and Myron Scholes proposed Black-Scholes model(Or Black-76 model) to obtain the price of European option theoritically.

In the Black-Scholes model, it was supposed that the asset volatility is constant. However, if we compute the implied volatility from market price, it has different values according to maturities and strike prices of the option. The former is called volatility term structure and the latter is called volatility smile(or skew). We can see above phenomena through the volatility surface.

To solve suggested problems, the local volatility model has been developed by Dupire(1994), Derman-Kani(1994). Dupire has considered calibration from market date to get local volatility function instead deriving formula theoritically. The local volatility model is one of the most successful for equity models. In practice, it is important to price and hedge derivatives under the local volatility model.

In this article, I introduce the Black-Scholes model, the local volatility model and method to obtain the price of European vanila option by solving Black-scholes PDE via finite differential.

Memo